Unit XVII: Enzymes

Lecture notes: the slides plus what she explained in class (transcript)

CHEMISTRY (PROF. L. RANAY)

WHY ENZYMES MATTER

- Catalysis is **the most important function of proteins**. Catalysis can be up to a **10^{20}-fold** increase (slide 3); enzyme-controlled reactions proceed **10^{8} to 10^{11} times faster** than the same reactions without enzymes (slide 21).
- Without enzymes, digesting one meal would take years: no energy production, so **no continuity of life**.
- Enzymes matter in **metabolism, diagnosis and therapeutics**:
  - **Diagnosis:** blood enzyme levels indicate disease. In **myocardial infarction (heart attack)**, **lactate dehydrogenase (LDH)** and **creatine kinase (CK)** rise; **CK shows up more rapidly than LDH**.
  - **Therapy:** e.g., **digestive enzymes**.

WHAT A REACTION NEEDS (HER REVIEW)

- **1. Enough activation energy** (some reactions need heating because their activation energy is high).
- **2. Proper orientation** (collision theory): molecules must collide properly so old bonds break and new bonds form.

DEFINITION AND CHARACTERISTICS

- **Enzymes are biological catalysts** that catalyze nearly all reactions in the cells. They are **proteins** that increase the rate by finding an **alternative pathway with a lower activation energy**.
- **Not altered or consumed** during the reaction; **reusable**.
- **Highly specific** (unlike inorganic catalysts).
- **Sensitive to their environment**, so they can be controlled by **temperature, pH or substrate concentration** (protein denaturation is covered in lab).
- **Structure: globular** shape with a complex 3-D structure (α-helices, β-sheets and side chains; example on the slide: human pancreatic amylase). Globular so they travel easily in the body's aqueous environment.

SIMPLE VS CONJUGATED ENZYMES

- **Simple:** protein only (only a few enzymes). **Conjugated:** protein part + non-protein part; **almost all** enzymes are conjugated.
- **Apoenzyme** = the protein part, the enzyme **without** its non-protein part; it is **inactive**.
- **Cofactor** = the non-protein part. **Apoenzyme + cofactor = holoenzyme** (the **active** enzyme).
- Enzymes are **highly regulated**: cofactors are one way to activate them.
- **Prosthetic group: tightly bound** to the apoenzyme; usually a small **inorganic** molecule or atom (e.g., **Fe**), may be organic (e.g., **FAD**); often **remains intact** during the reaction.
  - She: tightly bound means held by a **covalent bond** or a **salt bridge** (ionic). Metal cations (Fe^{2+}/Fe^{3+}, Zn^{2+}, Mg^{2+}) bind to **acidic side chains** (COO^- in neutral solution).
- **Coenzyme: loosely bound**; an **organic** molecule, often **vitamin-derived** (e.g., **CoA, NADH**).
  - She: loosely bound means held only by **intermolecular forces** (hydrogen bonds, London dispersion, dipole-dipole), easy to break.
- **Coenzymes are a type of cofactor.** Each enzyme has its own cofactor.
- She: **vitamins and minerals** (the nutrition terms) are needed because they are **cofactors** of our enzymes.
- Denaturing a holoenzyme breaks the **secondary and tertiary** structure (not the chain), so the cofactor is affected too; **oxidizing or reducing agents** can also break **disulfide bonds**.

KEY TERMS

- **Active site:** the area of the enzyme where the **substrate** attaches and the reaction happens; enzymes are very large proteins and the active site is only a **small region**.

- **Substrate:** the **reactant** in the enzyme-catalyzed reaction. The enzyme is only the catalyst, not the main player.
- The **primary structure** matters: the folding must bring the right residues (her example: a histidine at position 127) into the active site, or no catalysis happens.
- **Pepsinogen → pepsin:** pepsinogen (in the stomach) is **inactive** until it receives **HCl (H^+)**, then it becomes **pepsin**. Inactive forms often end in **-gen**, active forms in **-in**. (Hyperacidity medicines are proton inhibitors.)

NAMING ENZYMES

- Usually end in **-ase**: **sucrase** catalyzes the hydrolysis of **sucrose**.
- Some keep **common names**, especially digestive enzymes: **pepsin, trypsin**.
- The name can describe the **function** (**oxidase** catalyzes oxidation) or **both the substrate and the function** (**alcohol dehydrogenase** oxidizes ethanol).
- She: "I will not ask you to name" enzymes, but **remember the classification**.

CLASSIFICATION (EC NUMBERS)

- Classified by the **International Union of Biochemists (IUB)** by the **type of reaction** they catalyze; each enzyme has an **Enzyme Commission (EC) number** of **four numbers separated by periods**. The class number is fixed internationally.

SUMMARY TABLE: ENZYME CLASSES (EC NUMBERS)

EC	Class	Reaction catalyzed	Slide examples
1	**Oxidoreductases**	Oxidation-reduction reactions	glucose oxidase, peroxidase, catalase, phenylalanine hydroxylase
2	**Transferases**	Group transfer reactions	transaminases (ALT and AST), kinases (phosphotransferases), transmethylases, transpeptidases, transacylases
3	**Hydrolases**	Hydrolysis reactions; **water is the acceptor** of the transferred group	peptidases, amylase, maltase, lactase, lipase, deaminases, phosphatases
4	**Lyases**	Cleave bonds by means **other than hydrolysis and oxidation**; add or remove H_2O, NH_3 or CO_2 across double bonds	fumarase, carbonic anhydrase, hydratase, dehydratase
5	**Isomerases**	Isomerization reactions: L to D, mutase reactions (shifts of chemical groups)	isomerase, mutase
6	**Ligases**	**Join** two molecules with covalent bonds, using energy from **ATP**	acetyl-CoA carboxylase, glutamine synthetase

- **EC 7, translocases** (slide 18): move ions or molecules **across membranes**; maintain electrochemical gradients and cellular transport.

READING AN EC NUMBER: HEXOKINASE, EC 2.7.1.1

- **2** = class (**transferases**); **7** = subclass (transfer of a **phosphorus-containing group**); **1** = sub-subclass (phosphotransferase with an **alcohol group as acceptor**); **1** = serial number (the specific enzyme).
- Systematic name: **ATP:D-hexose-6-phosphotransferase**: it takes a **phosphate from ATP** and puts it on a **hexose**: glucose + ATP → **glucose-6-phosphate** + ADP (the first step of glucose metabolism).

HOW TO SPOT THE CLASS (HER "SPOT THE DIFFERENCE")

- **Oxidoreductase:** look at O and H: OH → C=O (H removed = oxidation), a new double bond, alcohol → aldehyde → acid, 2° alcohol → ketone.
- **Transferase:** a group moves from one molecule to another (ATP → ADP + phosphorylated sugar).
- **Hydrolase: water** breaks a big molecule into smaller ones.
- **Lyase:** a bond is cleaved **without** water or oxidation; a double bond appears or disappears.

- **Isomerase:** same atoms, only the **position** changes, or **L becomes D**.
- **Ligase:** two molecules joined, using **ATP**.

Slide exercise

(a) a C–OH becomes C=O / a double bond forms → **oxidoreductase**. (b) water is added to break the molecule → **hydrolase**.

HOW ENZYMES WORK

- Every reaction needs an initial input of energy, the **activation energy**; at the top the molecules are in the **transition state**.
- Biological systems are very sensitive to temperature, so enzymes speed up reactions **without raising the temperature**: they **lower the activation energy** by finding a **new pathway ("a short cut")**. They do not remove the activation energy (her analogy: going around the mountain instead of cutting it down).
- The enzyme also gives the **highest chance of proper collision**: the active site holds the reactants in the right **orientation**. So the enzyme handles **both** requirements: lower activation energy **and** proper orientation.
- **Enzyme + substrate ⇌ enzyme-substrate (ES) complex → enzyme + product.** The ES complex is the **intermediate**.

LOCK-AND-KEY VS INDUCED FIT

- **Lock-and-key:** the active site is **rigid**; only a substrate with the **matching shape** fits (substrate = key, active site = lock). Her example: **sucrase** acts only on **sucrose** (one to one).
- **Induced fit:** the active site is **flexible** and changes shape; enzyme and substrate adjust to **maximize the fit**; **greater range of substrate specificity**; fits a **wider range of enzymes** (most enzymes). Still specific: **alcohol dehydrogenase** accepts methanol, ethanol, isopropyl alcohol (all have the OH) but not non-alcohols.

SPECIFICITY

SUMMARY TABLE: TYPES OF ENZYME SPECIFICITY

Type	What it means	Example
Absolute	One reaction for **one specific substrate**	**urease** hydrolyzes only urea; **sucrase** acts only on sucrose
Group	One reaction for substrates with **similar functional groups**	**hexokinase** adds phosphate to hexoses (glucose, fructose, mannose); **alcohol dehydrogenase** acts on a range of alcohols
Linkage	One type of reaction for a specific **type of bond**	**salivary amylase** hydrolyzes α-1,4-glycosidic bonds in starch; **pepsin** cuts peptide bonds next to aromatic or hydrophobic residues
Stereospecificity	Acts on one particular **stereoisomer** (chirality is inherent in the active site)	**L-amino acid oxidase** oxidizes L-amino acids but not D-amino acids

- She: **pepsin** breaks peptide bonds, but only those next to **aromatic or hydrophobic (nonpolar)** amino acids, not every peptide bond.
- She: **starch** (α-glycosidic bonds) is digested, but **cellulose** (**β**-glycosidic bonds, also glucose units) is not: we have no enzyme for the β bond (why leafy vegetables like kangkong pass through).
- She: an **L-glucose** IV would be treated as foreign/toxic: our enzymes act on **D-sugars** and **L-amino acids**.

FACTORS AFFECTING ENZYME ACTIVITY

- **Environmental/physicochemical:** temperature, pH, substrate concentration, enzyme concentration.
- **Regulatory molecules:** activators (cofactors, allosteric activators) and inhibitors (competitive, noncompetitive, uncompetitive, irreversible, allosteric).

TEMPERATURE

- Activity **rises up to the optimum temperature** (where the reaction is fastest), then falls.
- Most enzymes: optimum about **40 °C**. Many are much lower: **cold-water fish die at 30 °C** because their enzymes denature. A few **bacteria** have enzymes that withstand up to **100 °C**. Most enzymes are **fully denatured at 70 °C**.

PH

- A pH slightly off the optimum causes small changes in the **charges** of the enzyme and substrate; this change in ionization affects **substrate binding** at the active site.
- Each enzyme has its own **optimum pH**; **extreme pH** causes **denaturation**.

SUBSTRATE CONCENTRATION

- At constant enzyme concentration, the rate **increases** with substrate concentration until the enzyme is **saturated** (all enzymes bound to substrate) = **saturation point**, maximum activity (**Vmax**).
- Changing the **enzyme** concentration changes **Vmax**.

ENZYME CONCENTRATION

- The rate **increases** with enzyme concentration, assuming the enzyme concentration is **much lower** than the substrate's.

COFACTORS AND COENZYMES (DETAILS)

- A cofactor is an additional **non-protein** molecule some enzymes need to work properly; its role is to **assist**. Tightly bound = **prosthetic group**; bound and released easily = **coenzyme**.
- **Many vitamins are coenzymes**; they are **not specific** to which enzyme they assist. Slide picture: **nitrogenase** with Fe, Mo and ADP cofactors.

METAL COFACTORS

- **Cu^{2+}:** cytochrome oxidase (last enzyme of the respiratory electron transport chain).
- **Fe^{2+}/Fe^{3+}:** catalase (hydrogen peroxide → water + oxygen).
- **Mg^{2+}:** hexokinase (phosphate to glucose → glucose-6-phosphate, glycolysis).
- **Mn^{2+}:** arginase (final enzyme of the urea cycle).
- **Zn^{2+}:** alcohol dehydrogenase, with NAD^+ (alcohol → aldehyde → carboxylic acid).

COENZYMES AND THEIR VITAMINS

- **FAD/$FADH_2$ (vitamin B2), flavin adenine dinucleotide:** reducing agent in cellular respiration; donates electrons to the electron transport chain.
- **NAD/NADH (vitamin B3), nicotinamide adenine dinucleotide:** carries and transfers electrons; oxidizing agent in redox reactions.
- She: **NADH and $FADH_2$ are electron carriers** that feed the last part of glucose metabolism, which generates **ATP**.
- **CoA (from vitamin B5): acyl group carrier**. She: acetyl-CoA starts the **Krebs cycle**; the acetyl group cannot enter the mitochondria without CoA.
- **Coenzyme Q10 (ubiquinone):** oil-soluble, vitamin-like; carries electrons in aerobic cellular respiration. **Quinones:** conjugated aromatic rings; pigments in bacteria, fungi, some plants.
- **Vitamin C (ascorbic acid):** co-substrate with iron in **collagen** formation; helps vitamins E and B9 stay active.
- **TPP (B1, thiamin pyrophosphate):** citric acid cycle, transfer of a carboxyl group. **B6 compounds:** transfer of **amino** groups. **Biotin (B7/vitamin H):** transfer of a **carboxyl** group. **THF (B9, tetrahydrofolate): methylation**. **Cobalamin (B12):** transfer of a **methyl** group. Also vitamins A and K.

INHIBITORS

- Substances that **reduce the rate** of enzymatic reactions; usually **specific**, work at **low concentrations**; **block** the enzyme but usually do not destroy it. Many **drugs and poisons** inhibit enzymes in the nervous system. **Reversible or irreversible.**

SUMMARY TABLE: TYPES OF ENZYME INHIBITION

Type	Binds where	Resembles the substrate?	Effect of more substrate	Reversible?
Competitive	The **active site** (competes with the substrate)	**Yes**, closely	**Reverses** the inhibition	Reversible
Noncompetitive	**Not the active site** (allosteric site); distorts the enzyme's shape	No (different structure)	**Not influenced** by substrate concentration	Reversibly or irreversibly
Uncompetitive	Only the **enzyme-substrate complex**, not at the active site	No (different structure)	Substrate **binds longer** in the enzyme and the rate drops	Slide says it binds irreversibly
Irreversible	Listed as a mode of action; not detailed on the slides	Not stated	Not stated	Irreversible

- **Competitive** (she): the inhibitor enters the active site so the substrate has no space; reversible: adding **more substrate** pushes it out. **Methanol poisoning** is treated with **ethanol**: ethanol competes with methanol for the active site (alcohol dehydrogenase), lessening the damage.
- **Noncompetitive** (she): binds at the **allosteric site** (not the active site), changes the enzyme's shape, so the substrate no longer fits; it can act **even before** the substrate binds.
- **Uncompetitive** (she): acts **only on the enzyme-substrate complex** (the substrate is already bound), also at an allosteric site.

APPLICATIONS OF INHIBITORS

- **Poisons:** snake bite, plant alkaloids, nerve gases. **Medicine:** antibiotics, sulfonamides, sedatives, stimulants.
- **Aspirin:** inhibits **prostaglandin** synthesis (aches and pains of arthritis).
- **Methotrexate:** cancer chemotherapy; semi-selectively inhibits **DNA synthesis** of malignant cells.
- **Sulfa drugs:** inhibit **folic acid synthesis** that bacteria need to grow.

ALLOSTERIC REGULATION

- **Allosteric** = binds at a site other than the active site. Not all enzymes are allosteric.
- **Negative regulator (inhibitor):** decreases the rate; changes the active site so the substrate is less readily accepted.
- **Positive regulator (activator):** increases the rate; reshapes the active site so it accepts substrate more readily. Example: **fructose-2,6-bisphosphate activates phosphofructokinase-1**, increasing glycolysis in response to **insulin**. Also listed: hexokinase-1, glucokinase.
- Slide's allosteric enzyme example: inhibitor **cytidine triphosphate (CTP)**, activator **adenosine triphosphate (ATP)**.

FEEDBACK CONTROL

- **Activation or inhibition of the first reaction** in a sequence is controlled by a **product of the sequence** (usually the final product inhibits the first step). Reactions in the body come in **series**, and the series itself is how they are controlled.

WHAT TO REVIEW

- Characteristics; simple vs conjugated; apoenzyme + cofactor = holoenzyme; prosthetic group (tight, covalent/salt bridge) vs coenzyme (loose, IMF, vitamin).
- Active site, substrate, ES complex; lowering activation energy + orientation; transition state.
- The six (seven) EC classes with how to spot each; hexokinase EC 2.7.1.1.
- Lock-and-key vs induced fit; four specificity types with examples (urease, hexokinase/ADH, salivary amylase, L-amino acid oxidase).
- Temperature, pH, substrate and enzyme concentration effects (40 °C, 70 °C, saturation, Vmax).
- Metal cofactors and coenzyme-vitamin pairs; $NADH/FADH_2$ electron carriers; CoA.

- Competitive vs noncompetitive vs uncompetitive; methanol + ethanol; aspirin, methotrexate, sulfa drugs; allosteric regulators; feedback control; LDH and CK in heart attack.